

## SJF vs Priority Comparison Project

### Cover Sheet - Algorithm Comparison Project

|                       |                                                                                                               |                    |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------|--------------------|-----|
| <b>Project Title</b>  | SJF vs Priority Comparison Project                                                                            | <b>Team Number</b> | 120 |
| <b>Repository URL</b> | <a href="https://github.com/saraomar-dev/our_os_projectt">https://github.com/saraomar-dev/our_os_projectt</a> |                    |     |

### Team Members

| No. | Student Name                | Student ID | Contribution Area | Grade | Notes |
|-----|-----------------------------|------------|-------------------|-------|-------|
| 1   | ساره عمر محمد فتح الله      | 20240404   |                   |       |       |
| 2   | سما محمد شعبان احمد         | 20240439   |                   |       |       |
| 3   | اريج وائل عبدالرؤف صالح     | 20240117   |                   |       |       |
| 4   | سجده إبراهيم سليمان إبراهيم | 20240413   |                   |       |       |
| 5   | روان عمرو محمد شحاته        | 20240364   |                   |       |       |
| 6   | رنا محمد سعد محمد بكر       | 20240357   |                   |       |       |
| 7   |                             |            |                   |       |       |

### Evaluation Rubric Summary

| Criterion                                | Marks | Specific focus      | Notes for evaluator                                                                                   |
|------------------------------------------|-------|---------------------|-------------------------------------------------------------------------------------------------------|
| Interface and usability                  | 15    | -                   | Clean layout, easy data entry, easy comparison workflow, readable output.                             |
| Input handling and validation            | 10    | -                   | Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.         |
| Algorithm A correctness                  | 15    | SJF                 | Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules. |
| Algorithm B correctness                  | 15    | Priority Scheduling | Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules. |
| Gantt charts and execution visualization | 10    | -                   | Separate readable Gantt chart for each algorithm with correct time markers and execution order.       |

|                                  |    |                 |                                                                                             |
|----------------------------------|----|-----------------|---------------------------------------------------------------------------------------------|
| Metrics calculation              | 10 | -               | Correct WT, TAT, RT for each process and correct averages, consistent with charts.          |
| Comparison quality and fairness  | 10 | Both algorithms | Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.  |
| Required test scenarios          | 5  | -               | Normal case, behavior-revealing case, and invalid-input validation case are all included.   |
| Report and technical explanation | 5  | -               | Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions. |
| Viva / team understanding        | 5  | -               | Team members can explain both algorithms, outputs, metrics, and design decisions.           |

### Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

| Check                    | Checklist Item                                                                   | Notes |
|--------------------------|----------------------------------------------------------------------------------|-------|
| <input type="checkbox"/> | SJF selection is based on the shortest available burst time.                     |       |
| <input type="checkbox"/> | Priority rule is stated clearly and applied consistently.                        |       |
| <input type="checkbox"/> | At least one workload shows conflict between short burst time and high priority. |       |
| <input type="checkbox"/> | Comparison tables use the same dataset for both algorithms.                      |       |
| <input type="checkbox"/> | Conclusion explains efficiency versus urgency trade-off.                         |       |

### Project-Specific Notes for Students

- Use the same workload in both algorithms.
- Assume **preemptive** SJF and Priority Scheduling.
- Discuss efficiency versus urgency when short jobs and high-priority jobs conflict.
- Explain whether either policy causes unfair delay.

## Detailed Submission Description

The following requirements apply to all comparison projects.

### Implementation Technology

- Programming language: **any programming language**.
- GUI technology: must be used for the interface and visualization components.

### GitHub Submission Guidelines

- Repository: the full project must be submitted through GitHub in a clean repository with meaningful commits.
- The repository must include source code, README, screenshots, sample test cases, and run instructions.
- The repository should include a clear package structure and separate classes for models, scheduling logic, GUI, metrics, and utilities.

### Project Submission Form Guidelines

- A Project Submission Form must be completed and submitted with the GitHub repository link, project title, team number, and student information.
- The default branch must build and run correctly at submission time.

### Recommended Repository Structure

```
project-root/
  src/
    model/
    scheduler/
    metrics/
    gui/
    util/
  assets/
  screenshots/
  test-cases/
  README.md
  .gitignore
```

### Submission Checklist with Notes

| Done                     | Submission Item                           | Notes                                                                                   |
|--------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------|
| <input type="checkbox"/> | GitHub repository link provided           | The submitted link must be accessible and point to the final repository.                |
| <input type="checkbox"/> | README included                           | Must contain project description, requirements, build/run steps, and team member names. |
| <input type="checkbox"/> | Source code included                      | Do not submit compiled files only. Source code must be complete.                        |
| <input type="checkbox"/> | Executable/runnable instructions included | List Java version, JavaFX setup, IDE or command-line execution steps.                   |
| <input type="checkbox"/> | Screenshots included                      | Include interface screenshots and at least one Gantt chart screenshot per algorithm.    |
| <input type="checkbox"/> | Test scenarios included                   | Provide at least three documented test scenarios with outputs.                          |

|                          |                                   |                                                         |
|--------------------------|-----------------------------------|---------------------------------------------------------|
| <input type="checkbox"/> | Project Submission Form completed | Must match the repository and team information exactly. |
|--------------------------|-----------------------------------|---------------------------------------------------------|